

NORMALIZATION

Introduction to RDBM – Mini Project 1

1. Normalize the following table into 1 Normal form:

Member_Id	First_Name	Last_Name	Hobbies
101	Jayson	Mark	Cricket,Swimming,Football
102	Ram	Ganesh	Swimming,Running,Music
103	Raj	Kishore	Dancing, Singing, Running

```
SQL*Plus: Release 21.0.0.0.0 - Production on Wed Aug 5 22:16:26 2026
Version 21.3.0.0.0
```

```
Copyright (c) 1982, 2021, Oracle. All rights reserved.
```

```
Enter user-name: system
```

```
Enter password:
```

```
Last Successful login time: Wed Aug 05 2026 22:05:33 +05:30
```

```
Connected to:
```

```
Oracle Database 21c Express Edition Release 21.0.0.0.0 - Production
Version 21.3.0.0.0
```

```
SQL> CREATE TABLE MEMBER_1NF (
  2     MEMBER_ID NUMBER,
  3     FIRST_NAME VARCHAR2(20),
  4     LAST_NAME VARCHAR2(20),
  5     HOBBY VARCHAR2(30)
  6 );
```

```
CREATE TABLE MEMBER_1NF (
      *
```

```
ERROR at line 1:
```

```
ORA-00955: name is already used by an existing object
```

```
SQL> INSERT INTO MEMBER_1NF VALUES (101,'Jayson','Mark','Cricket');
```

```
1 row created.
```

```
SQL> INSERT INTO MEMBER_1NF VALUES (101,'Jayson','Mark','Swimming');
```

```
1 row created.
```

```
SQL> INSERT INTO MEMBER_1NF VALUES (101,'Jayson','Mark','Football');
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO MEMBER_1NF VALUES (102,'Ram','Ganesh','Swimming');
```

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```
1 row created.

SQL> INSERT INTO MEMBER_1NF VALUES (102,'Ram','Ganesh','Running');

1 row created.

SQL> INSERT INTO MEMBER_1NF VALUES (102,'Ram','Ganesh','Music');

1 row created.

SQL>
SQL> INSERT INTO MEMBER_1NF VALUES (103,'Raj','Kishore','Dancing');

1 row created.

SQL> INSERT INTO MEMBER_1NF VALUES (103,'Raj','Kishore','Singing');

1 row created.

SQL> INSERT INTO MEMBER_1NF VALUES (103,'Raj','Kishore','Running');

1 row created.

SQL>
SQL> COMMIT;

Commit complete.
```

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```
SQL> SELECT * FROM MEMBER_1NF;
```

MEMBER_ID	FIRST_NAME	LAST_NAME
HOBBY		
101	Jayson	Mark
Cricket		
101	Jayson	Mark
Swimming		
101	Jayson	Mark
Football		

MEMBER_ID	FIRST_NAME	LAST_NAME
HOBBY		
102	Ram	Ganesh
Swimming		
102	Ram	Ganesh
Running		
102	Ram	Ganesh
Music		

MEMBER_ID	FIRST_NAME	LAST_NAME
HOBBY		
103	Raj	Kishore
Dancing		
103	Raj	Kishore
Singing		

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```
      103 Raj
Running      Kishore
```

```
MEMBER_ID FIRST_NAME      LAST_NAME
-----
```

```
HOBBY
-----
```

```
      101 Jayson      Mark
Cricket
```

```
      101 Jayson      Mark
Swimming
```

```
      101 Jayson      Mark
Football
```

```
MEMBER_ID FIRST_NAME      LAST_NAME
-----
```

```
HOBBY
-----
```

```
      102 Ram      Ganesh
Swimming
```

```
      102 Ram      Ganesh
Running
```

```
      102 Ram      Ganesh
Music
```

```
MEMBER_ID FIRST_NAME      LAST_NAME
-----
```

```
HOBBY
-----
```

```
      103 Raj      Kishore
Dancing
```

```
      103 Raj      Kishore
```

```
      103 Raj      Kishore
Running
```

```
18 rows selected.
```

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2. Normalize the following Table into 2nd Normal Form:

Empno	Training	Training_Date	Dept
101	Oracle SQL	12-Aug-2015	TT
101	Java	21-Aug-2015	BU
102	Oracle SQL	18-Sep-2014	TT

```
SQL> CREATE TABLE EMPLOYEE (  
  2     EMPNO NUMBER PRIMARY KEY,  
  3     DEPT VARCHAR2(10)  
  4 );  
CREATE TABLE EMPLOYEE (  
  *  
ERROR at line 1:  
ORA-00955: name is already used by an existing object  
  
SQL> INSERT INTO EMPLOYEE VALUES (101,'TT');  
INSERT INTO EMPLOYEE VALUES (101,'TT')  
*  
ERROR at line 1:  
ORA-00001: unique constraint (SYSTEM.SYS_C008316) violated  
  
SQL> INSERT INTO EMPLOYEE VALUES (102,'TT');  
INSERT INTO EMPLOYEE VALUES (102,'TT')  
*  
ERROR at line 1:  
ORA-00001: unique constraint (SYSTEM.SYS_C008316) violated  
  
SQL>  
SQL> COMMIT;  
  
Commit complete.  
  
SQL> CREATE TABLE TRAINING (  
  2     EMPNO NUMBER,  
  3     TRAINING VARCHAR2(30),  
  4     TRAINING_DATE DATE,  
  5     FOREIGN KEY (EMPNO) REFERENCES EMPLOYEE(EMPNO)  
  6 );  
CREATE TABLE TRAINING (  
  *  
ERROR at line 1:  
ORA-00955: name is already used by an existing object
```

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```
SQL> INSERT INTO TRAINING VALUES (101,'Oracle SQL',TO_DATE('12-AUG-2015','DD-MON-YYYY'));
1 row created.

SQL> INSERT INTO TRAINING VALUES (101,'Java',TO_DATE('21-AUG-2015','DD-MON-YYYY'));
1 row created.

SQL> INSERT INTO TRAINING VALUES (102,'Oracle SQL',TO_DATE('18-SEP-2014','DD-MON-YYYY'));
1 row created.

SQL>
SQL> COMMIT;

Commit complete.

SQL> SELECT * FROM EMPLOYEE;

  EMPNO DEPT
-----
    101 TT
    102 TT

SQL>
SQL> SELECT * FROM TRAINING;

  EMPNO TRAINING              TRAINING_
-----
    101 Oracle SQL          12-AUG-15
    101 Java                21-AUG-15
    102 Oracle SQL          18-SEP-14
    101 Oracle SQL          12-AUG-15
    101 Java                21-AUG-15
    102 Oracle SQL          18-SEP-14

6 rows selected.
```

3. Normalize the following into 3rd Normal Form:

Member_Id	First_Name	Last_Name	Sports	Fees
101	Rajesh	Chand	Cricket	100
102	Jayesh	Raj	Hockey	80
103	Mark	Dorson	Football	90

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```
SQL> CREATE TABLE MEMBER (  
2     MEMBER_ID NUMBER PRIMARY KEY,  
3     FIRST_NAME VARCHAR2(20),  
4     LAST_NAME VARCHAR2(20)  
5 );  
CREATE TABLE MEMBER (  
      *  
ERROR at line 1:  
ORA-00955: name is already used by an existing object  
  
SQL> INSERT INTO MEMBER VALUES (101,'Rajesh','Chand');  
INSERT INTO MEMBER VALUES (101,'Rajesh','Chand')  
*  
ERROR at line 1:  
ORA-00001: unique constraint (SYSTEM.SYS_C008318) violated  
  
SQL> INSERT INTO MEMBER VALUES (102,'Jayesh','Raj');  
INSERT INTO MEMBER VALUES (102,'Jayesh','Raj')  
*  
ERROR at line 1:  
ORA-00001: unique constraint (SYSTEM.SYS_C008318) violated  
  
SQL> INSERT INTO MEMBER VALUES (103,'Mark','Dorson');  
INSERT INTO MEMBER VALUES (103,'Mark','Dorson')  
*  
ERROR at line 1:  
ORA-00001: unique constraint (SYSTEM.SYS_C008318) violated  
  
SQL>  
SQL> COMMIT;  
  
Commit complete.
```

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```
SQL> CREATE TABLE SPORTS (  
  2     SPORTS_NAME VARCHAR2(20) PRIMARY KEY,  
  3     FEES NUMBER  
  4 );  
CREATE TABLE SPORTS (  
      *  
ERROR at line 1:  
ORA-00955: name is already used by an existing object  
  
SQL> INSERT INTO SPORTS VALUES ('Cricket',100);  
INSERT INTO SPORTS VALUES ('Cricket',100)  
*  
ERROR at line 1:  
ORA-00001: unique constraint (SYSTEM.SYS_C008319) violated  
  
SQL> INSERT INTO SPORTS VALUES ('Hockey',80);  
INSERT INTO SPORTS VALUES ('Hockey',80)  
*  
ERROR at line 1:  
ORA-00001: unique constraint (SYSTEM.SYS_C008319) violated  
  
SQL> INSERT INTO SPORTS VALUES ('Football',90);  
INSERT INTO SPORTS VALUES ('Football',90)  
*  
ERROR at line 1:  
ORA-00001: unique constraint (SYSTEM.SYS_C008319) violated  
  
SQL>  
SQL> COMMIT;  
  
Commit complete.
```


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```
SQL> CREATE TABLE MEMBER_SPORTS (  
2     MEMBER_ID NUMBER,  
3     SPORTS_NAME VARCHAR2(20),  
4     FOREIGN KEY (MEMBER_ID) REFERENCES MEMBER(MEMBER_ID),  
5     FOREIGN KEY (SPORTS_NAME) REFERENCES SPORTS(SPORTS_NAME)  
6 );
```

```
CREATE TABLE MEMBER_SPORTS (  
      *
```

ERROR at line 1:

ORA-00955: name is already used by an existing object

```
SQL> SELECT * FROM MEMBER;
```

MEMBER_ID	FIRST_NAME	LAST_NAME
101	Rajesh	Chand
102	Jayesh	Raj
103	Mark	Dorson

```
SQL>
```

```
SQL> SELECT * FROM SPORTS;
```

SPORTS_NAME	FEES
Cricket	100
Hockey	80
Football	90

```
SQL>
```

```
SQL> SELECT * FROM MEMBER_SPORTS;
```

MEMBER_ID	SPORTS_NAME
101	Cricket
102	Hockey
103	Football